PERFECT OBJECTS ARE RELATIVELY PERFECT AND RELATIVE PERFECTNESS IS CLOSED UNDER DERIVED TENSOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Relative perfectness is checked on affine opens by the affine-local criterion. invoked at proof lines 9015-9016. Reduces both parts to affines.
- At the module level perfect complexes are relatively perfect and relative perfectness is closed under derived tensor with a perfect factor. invoked at proof lines 9017-9018. Concludes both parts after the reduction.

Source reference. Original lemma 0DI2: sources/stacks-perfect/pinned-original-source.tex, source lines 8951–8951. Cited at proof lines 9016-9016 for the reduction to affines; the module-level application is inside the counted span.

Source reference. Original lemma 0DHU: sources/stacks-chow/tags.txt, source lines 16919–16919. Cited at proof lines 9018-9018 to conclude both parts; the reduction is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes which is flat and locally of finite presentation. A perfect object of $D(\mathcal{O}_X)$ is S-perfect. If $K, M \in D(\mathcal{O}_X)$, then $K \otimes^{\mathbf{L}}_{\mathcal{O}_X} M$ is S-perfect if K is perfect and M is S-perfect.*

Proof. First proof: reduce to the affine case using Lemma 0DI2 and then apply More on Algebra, Lemma 0DHU. □

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.